



OSTEOPOROSIS DETECTION

Using Deep Learning Techniques

A detailed 3D rendering of a human knee joint, showing the femur, tibia, and patella in a light gray, semi-transparent style. The joint is positioned on the left side of the slide, with the knee bent at a 90-degree angle. The background is a solid dark gray.

PROPOSED OBJECTIVES

- To develop a deep learning-based system for automated osteoporosis detection using knee X-ray images.
- To apply transfer learning techniques for improving classification accuracy on limited medical datasets.
- To perform multi-class classification into normal, osteopenia, and osteoporosis categories.
- To address class imbalance issues using proper data handling techniques.
- To integrate explainable AI methods such as SHAP or Grad-CAM to interpret model decisions.
- To evaluate the model using clinical performance metrics like precision, recall, F1-score, and ROC-AUC.
- To design a user-friendly interface for real-time screening support.

These objectives ensure not only high accuracy but also fairness, interpretability, and clinical usability of the system

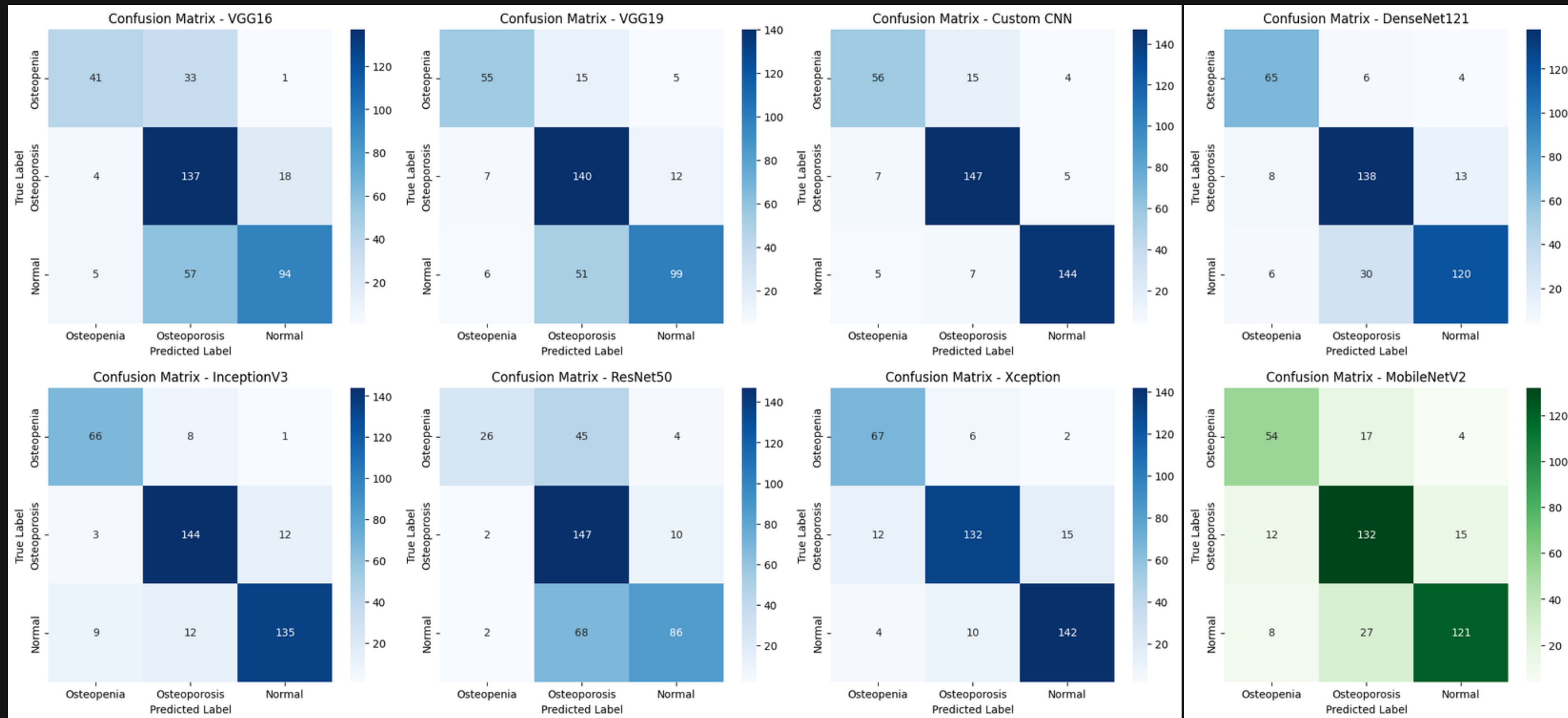
PROPOSED MODELS

- **Data Preprocessing**
 - All images were resized to a fixed input size required by CNN models
 - Normalization was applied to scale pixel values
- **Data augmentation techniques were used:**
 - Rotation
 - Horizontal flipping
 - Zooming
- **Augmentation helped in:**
 - Reducing overfitting
 - Handling class imbalance
 - Improving generalization

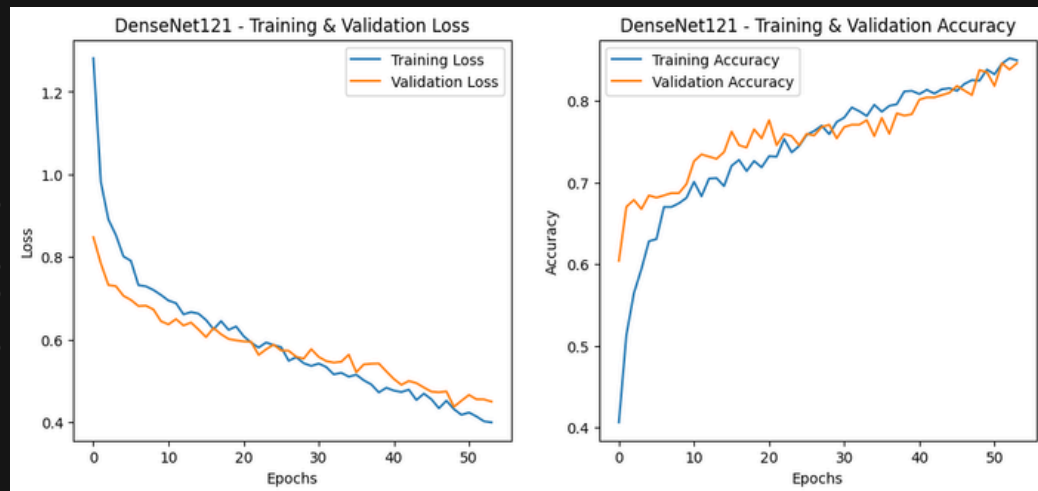
Model	Architecture Type	Key Strength	Computational Cost	Performance Trend
VGG16	Deep CNN (16 layers)	Simple & stable architecture	High	Good baseline accuracy
ResNet50	Residual Network	Handles deep networks efficiently using skip connections	Medium–High	Better than VGG
InceptionV3	Multi-scale CNN	Captures fine texture and spatial details	Medium	High accuracy
VGG19	Deep CNN (19 layers)	Rich feature extraction	High	Comparable / slightly better than VGG16
DenseNet121	Dense Connected Network	Reuses features from previous layers, improves gradient flow	Medium	Strong performance in medical imaging
MobileNetV2	Lightweight CNN	Designed for mobile & real-time applications	Low	Slightly lower but efficient accuracy
Xception	Depthwise Separable CNN	Efficient spatial feature extraction with fewer parameters	Medium	Near Inception-level accuracy
Custom CNN	Domain-specific CNN	Tailored for knee X-ray patterns	Low–Medium	Highest performance on our dataset

RESULTS

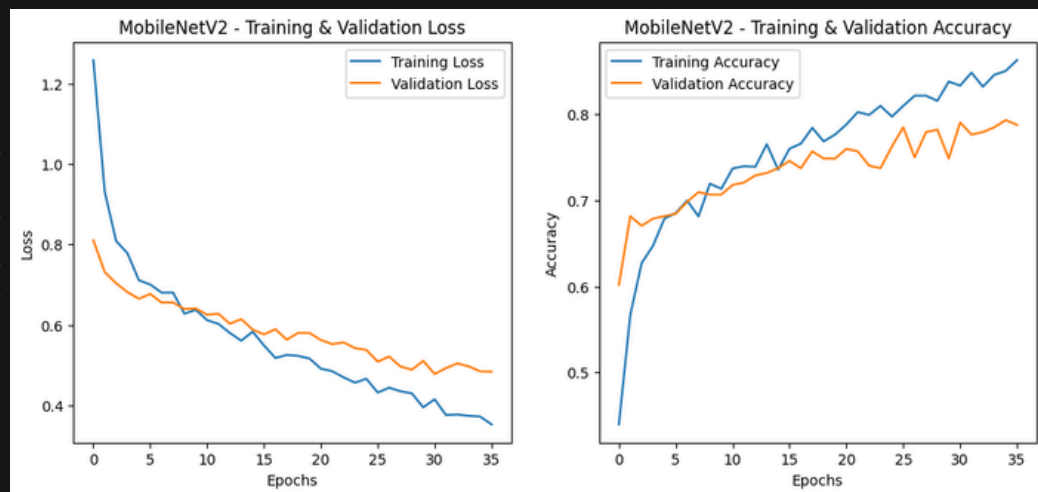
CONFUSION MATRICES



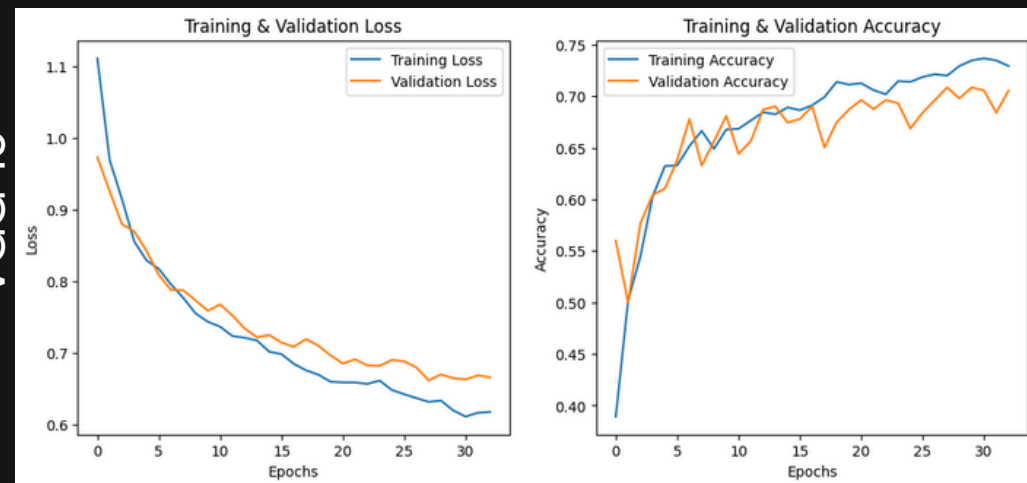
DenseNet121



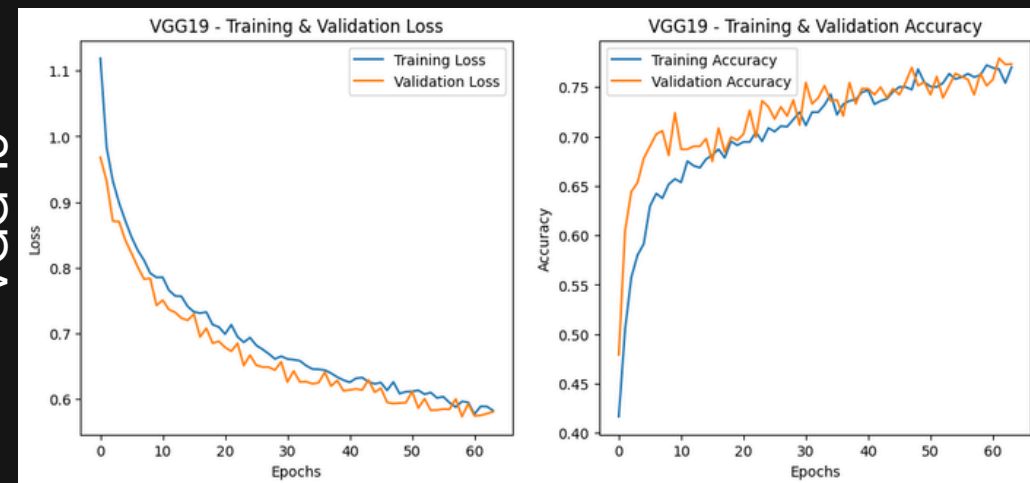
MobileNetV2



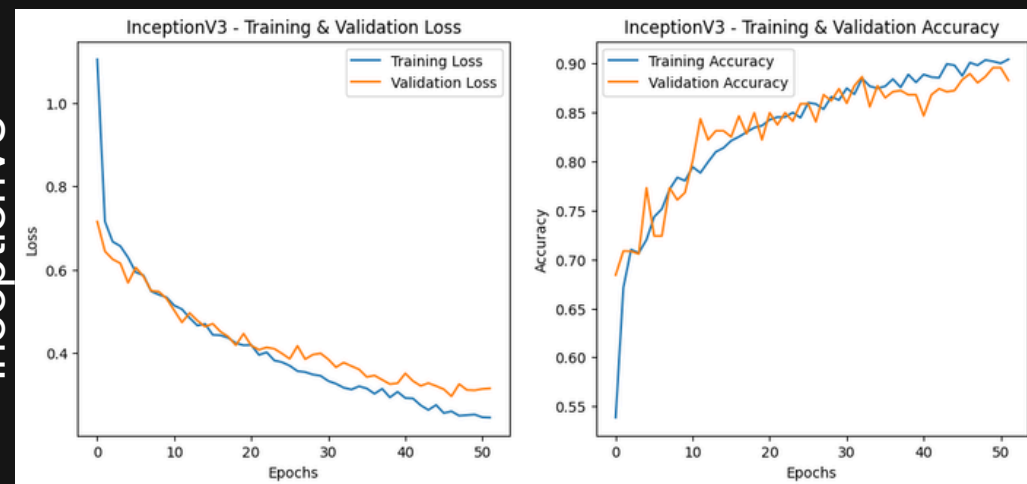
VGG 16



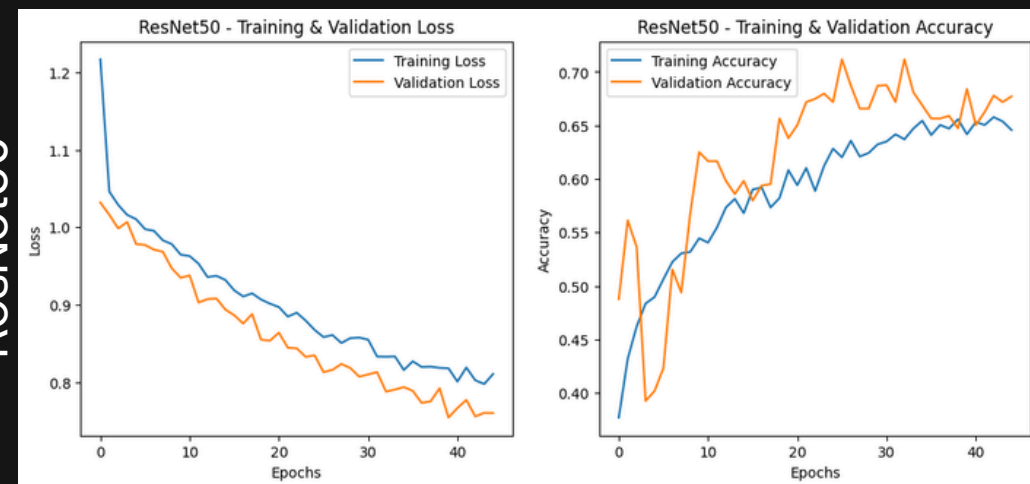
VGG 19



InceptionV3



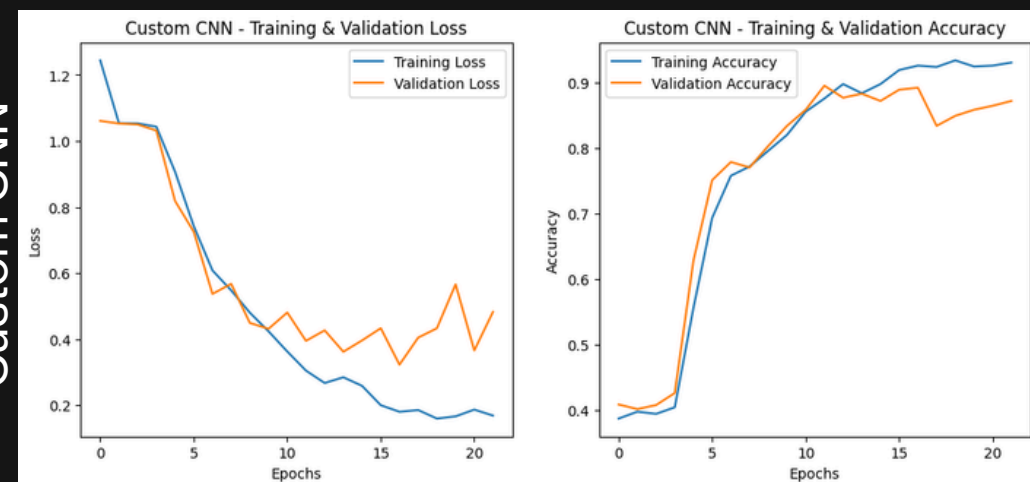
ResNet50



XceptionNet



Custom CNN



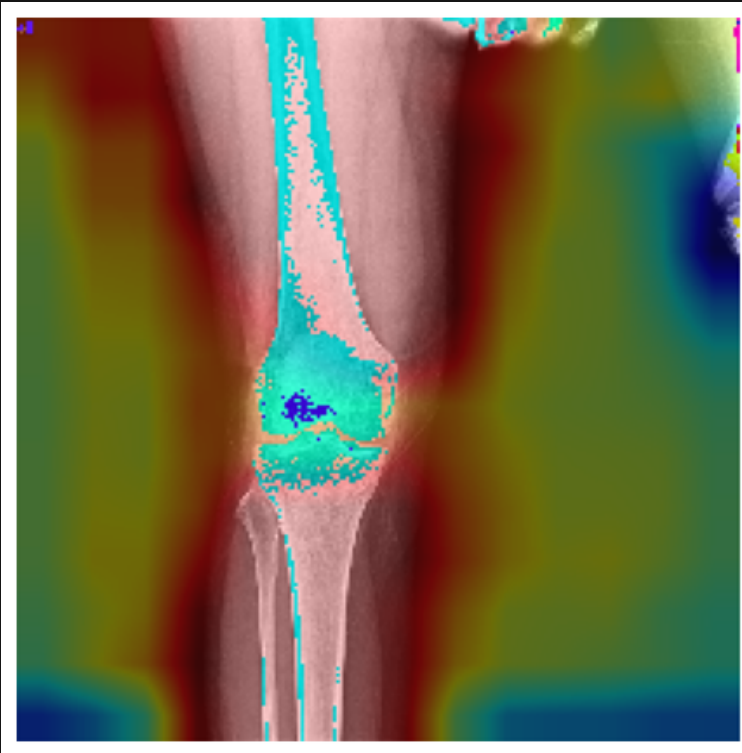
TRAINING GRAPHS

Model	Accuracy	Precision	Recall	F1-Score
VGG16	70%	0.74	0.7	0.7
VGG19	75%	0.77	0.75	0.75
ResNet50	66%	0.74	0.66	0.65
InceptionV3	88%	0.89	0.88	0.88
DenseNet121	83%	0.83	0.83	0.83
MobileNetV2	84%	0.82	0.84	0.82
Xception	87%	0.88	0.87	0.87
Custom CNN	89%	0.89	0.89	0.89

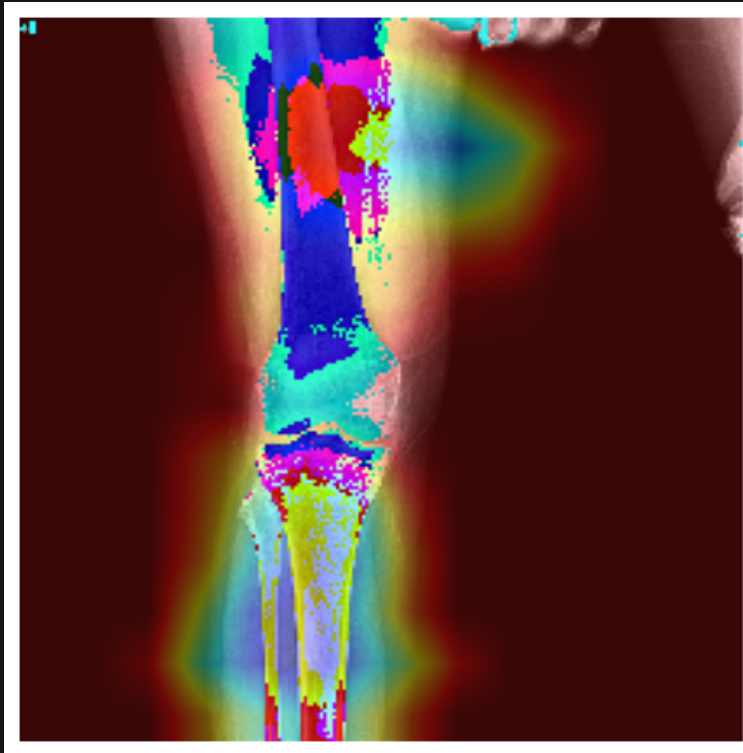
Table 2: Comparison of Results

GRAD-CAM RESULTS

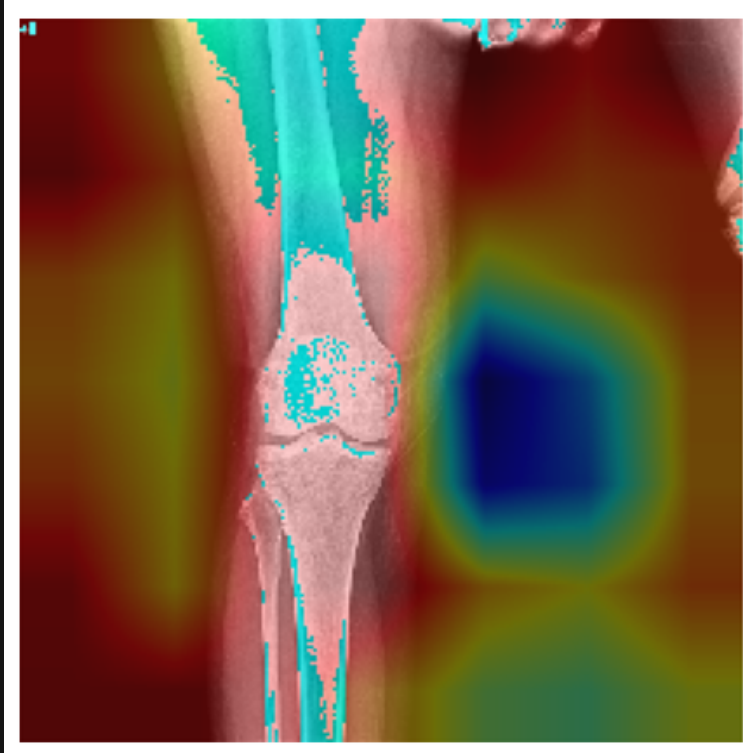
VGG 16



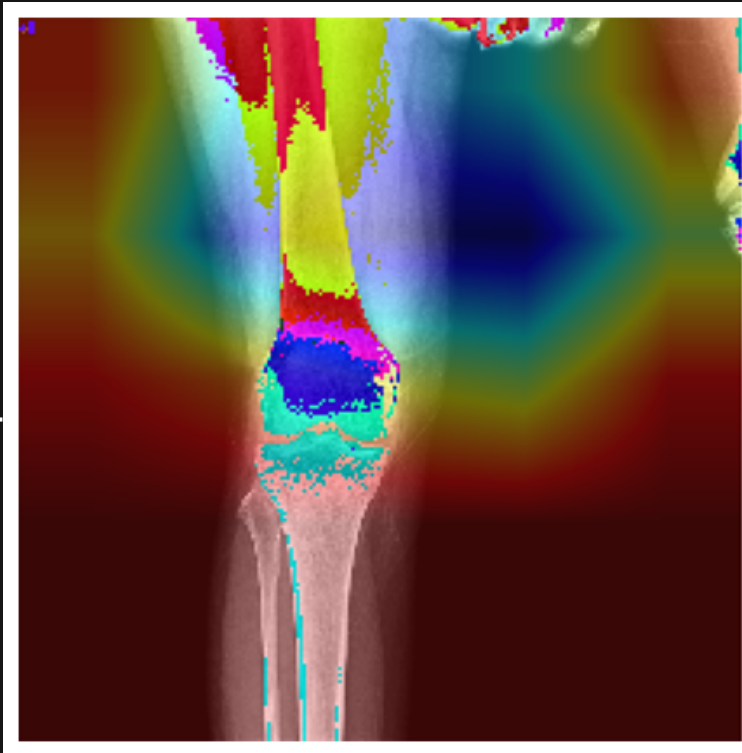
VGG 19



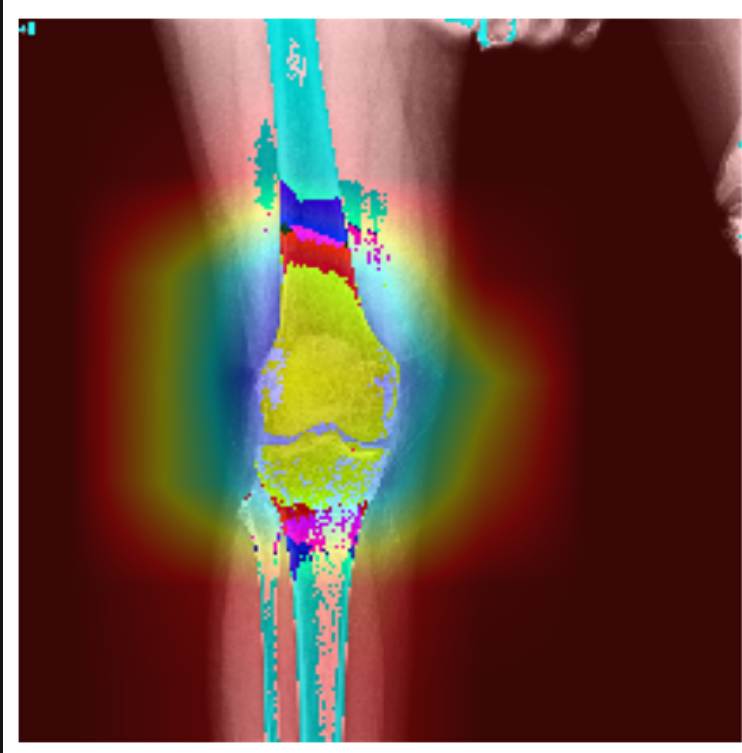
ResNet50



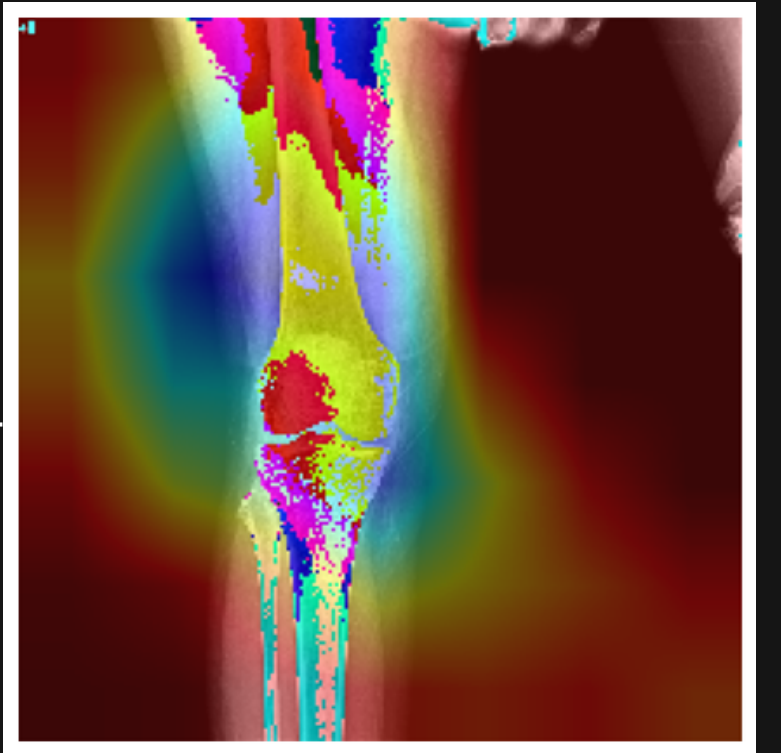
Inception V3



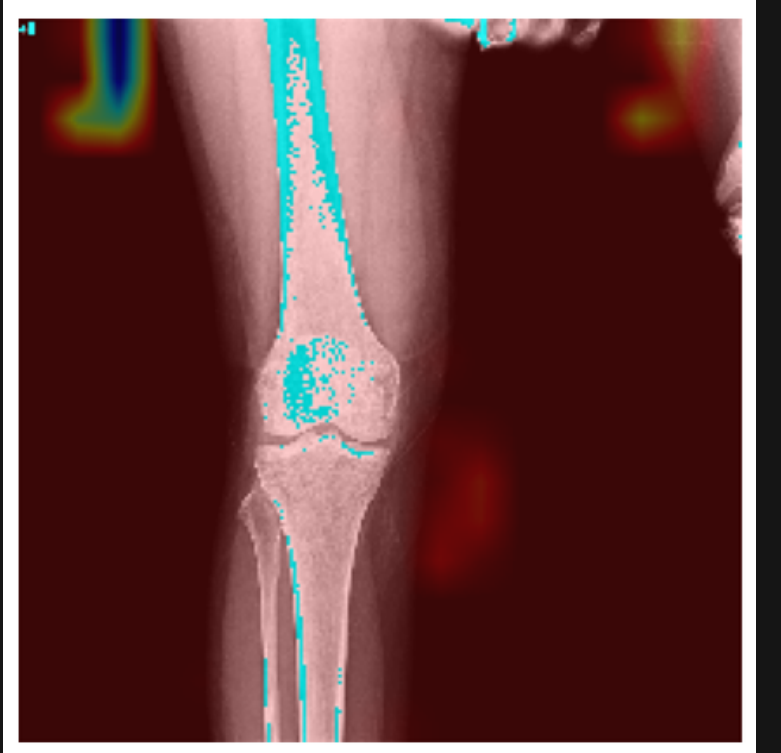
Densenet121



XceptionNet

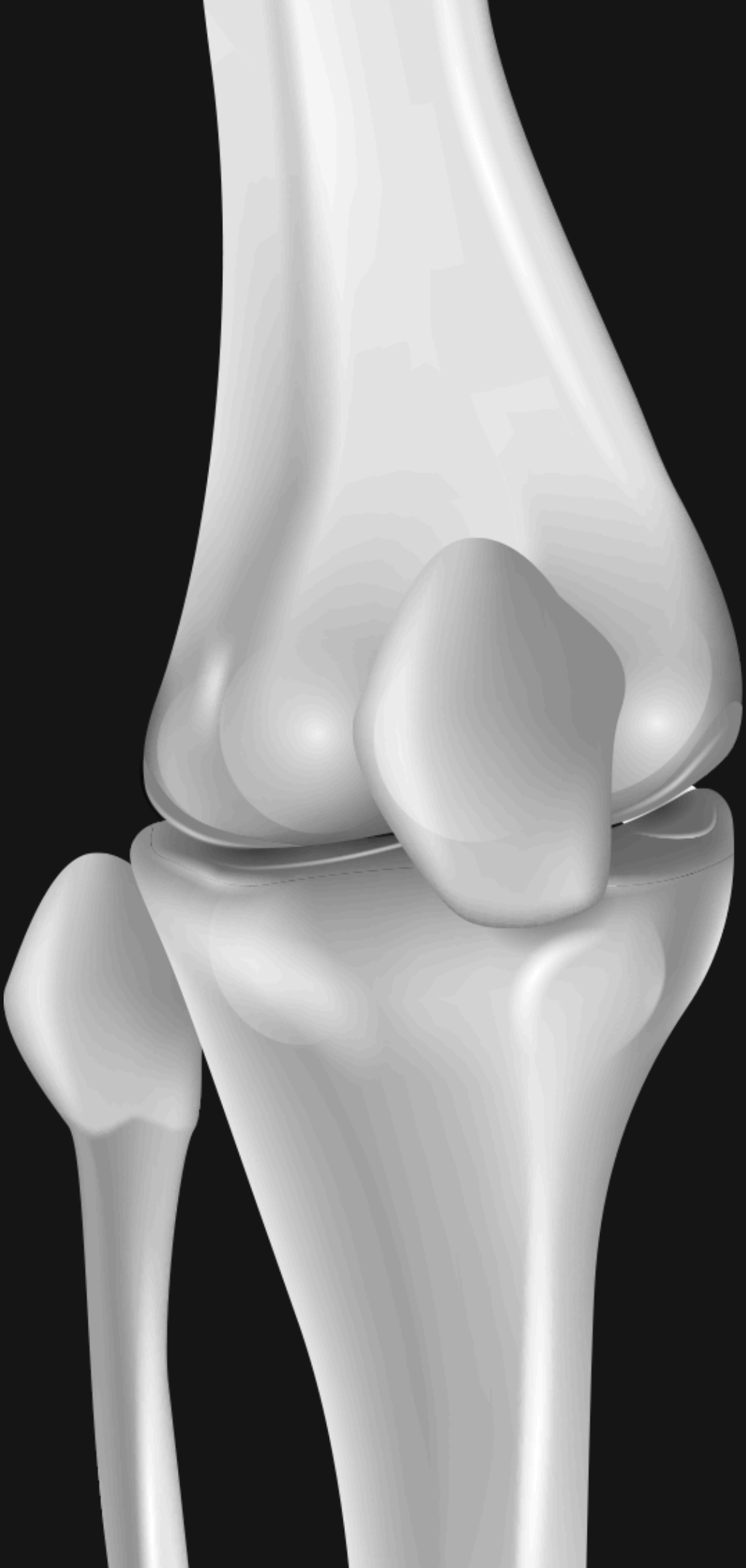
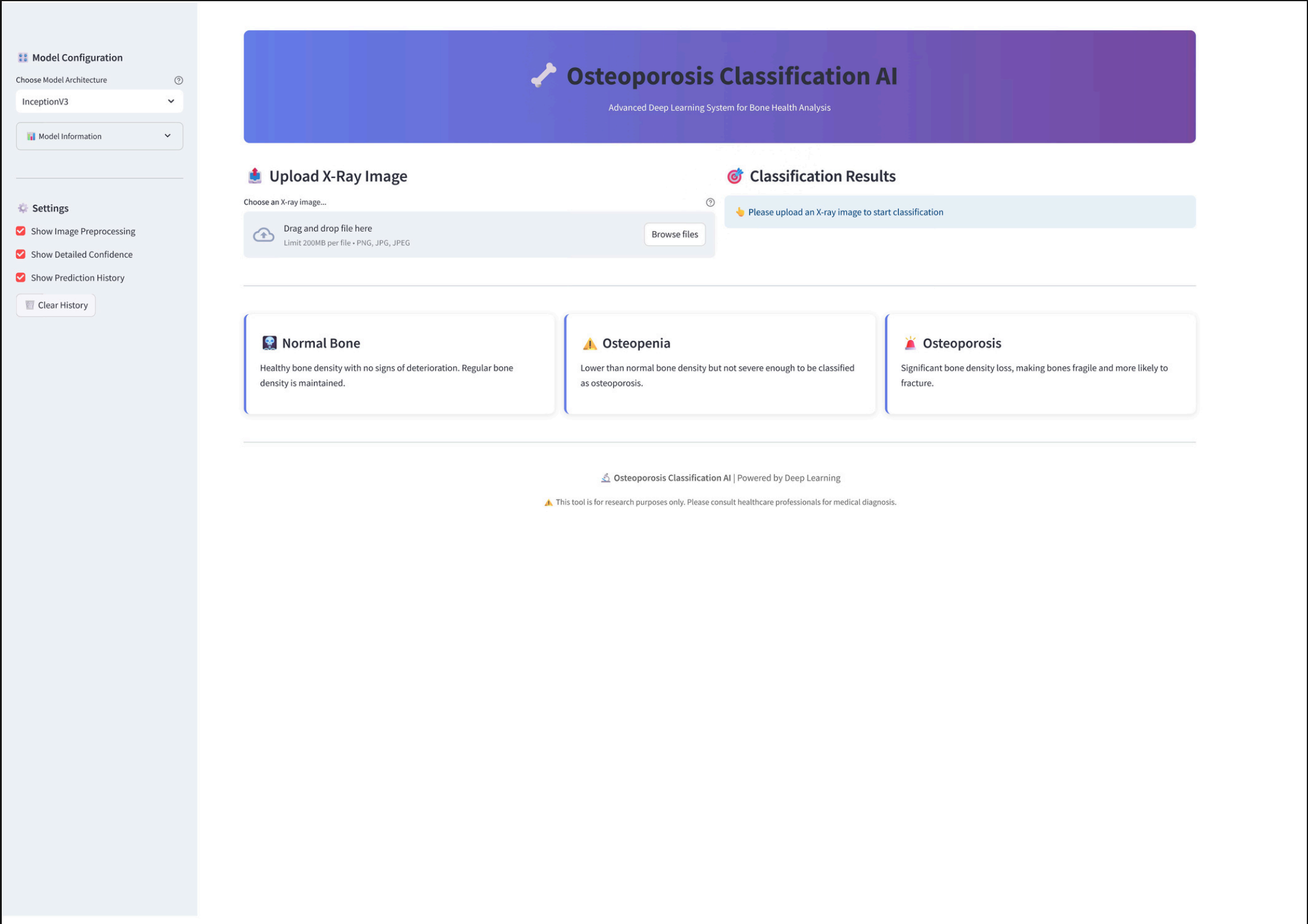


Custom CNN



- Most models focused on the knee joint region, which is clinically relevant for osteoporosis detection
- The custom CNN and Inception/Xception models concentrated more precisely on trabecular bone patterns, indicating better learning of bone texture features.

UI APPLICATION



TESTCASE VERIFICATION

Model Configuration

Choose Model Architecture

DenseNet121

Model Information

Settings

Show Image Preprocessing

Show Detailed Confidence

Show Prediction History

Clear History

Osteoporosis Classification AI

Advanced Deep Learning System for Bone Health Analysis

Upload X-Ray Image

Choose an X-ray image...

Drag and drop file here

Limit: 20MB per file • PNG, JPG, JPEG

Browse files

051.jpg 431.9KB



Classification Results

Diagnosis Result

Normal

Confidence: 86.86%

Model Used: DenseNet121

Timestamp: 2026-02-25 19:40:46

Confidence Breakdown

Normal

86.86%

Osteopenia

7.91%

Osteoporosis

5.22%

Model Configuration

Choose Model Architecture

DenseNet121

Model Information

Settings

Show Image Preprocessing

Show Detailed Confidence

Show Prediction History

Clear History

Image Details

Size: 1024 x 1024 pixels

Mode: RGB

Format: JPEG

Image Preprocessing

Detailed Analysis

Classification Confidence

Confidence Score

0.8686

0.0791

0.0522

Normal

Osteopenia

Osteoporosis

Bone Condition

Prediction Distribution

100%

Normal

Normal Bone

Healthy bone density with no signs of deterioration. Regular bone density is maintained.

Osteopenia

Lower than normal bone density but not severe enough to be classified as osteoporosis.

Osteoporosis

Significant bone density loss, making bones fragile and more likely to fracture.

Osteoporosis Classification AI | Powered by Deep Learning

This tool is for research purposes only. Please consult healthcare professionals for medical diagnosis.

THANK YOU